Speech Emotion Recognition with Domain Adaptation for Psychotherapy-Oriented Analysis

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***Abstract*—Speech Emotion Recognition (SER) seeks to automatically identify the emotional state of a speaker based on acoustic and temporal characteristics of speech signals. Despite substantial progress achieved through deep learning, SER systems often exhibit poor generalization when applied to data originating from unseen domains. This limitation is particularly critical in psychotherapy-oriented applications, where robustness across speakers, recording conditions, and speaking styles is essential.**

In this work, we investigate the problem of domain shift in speech emotion recognition and evaluate domain adaptation techniques to mitigate its effects. A lightweight Convolutional-Recurrent Neural Network (CRNN) baseline is trained on the IEMOCAP dataset and evaluated both in-domain and cross-domain on the RAVDESS dataset. Cross-domain evaluation reveals a severe performance degradation, highlighting the presence of strong domain shift. To address this issue, we extend the baseline architecture using Domain-Adversarial Neural Networks (DANN), which encourage the learning of domain-invariant feature representations through adversarial training.

Experimental results demonstrate that the proposed DANN-based approach significantly improves cross-domain performance while maintaining stable accuracy on the source domain. Additionally, we compare the lightweight CRNN-based model with a large self-supervised pre-trained speech model (wav2vec 2.0) under the same domain adaptation framework. The comparison highlights the trade-offs between computational cost, model complexity, and generalization performance. Overall, the results demonstrate that explicit domain adaptation mechanisms are essential for achieving robust cross-domain speech emotion recognition.

I. Introduction

Human speech conveys rich emotional information through prosodic, spectral, and temporal cues that extend beyond the lexical content of spoken words. Automatic Speech Emotion Recognition (SER) has therefore become an important research area within affective computing, enabling applications in human–computer interaction, intelligent assistants, and mental health support systems [4]. In psychotherapy-oriented contexts, analyzing emotional patterns in speech may assist clinicians by providing supportive insights into affective trends such as frustration, sadness, or emotional neutrality across sessions. Importantly, such systems are not intended to perform clinical diagnosis but rather to function as auxiliary analytical tools.

Recent advances in deep learning have significantly improved SER performance by learning discriminative representations directly from audio data. Convolutional Neural Networks (CNNs) operating on time–frequency representations such as log-Mel spectrograms have proven effective at capturing local spectral patterns [3], while Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are well-suited for modeling temporal dependencies in speech [12]. Hybrid architectures combining CNNs and RNNs have therefore become a common baseline in SER research [4].

Despite these advances, a major challenge remains unresolved: models trained on one dataset often fail to generalize to another. This phenomenon, commonly referred to as domain shift, arises due to differences in recording environments, speaker demographics, emotional expression styles, and annotation protocols [2]. As a result, high in-domain performance does not necessarily translate to robust real-world applicability.

This paper explicitly investigates domain shift in speech emotion recognition and evaluates domain adaptation techniques to mitigate its impact. We first establish a CRNN baseline trained on the IEMOCAP dataset [1] and demonstrate its failure to generalize to the RAVDESS dataset [2]. We then extend the baseline model using Domain-Adversarial Neural Networks (DANN), which promote domain-invariant feature learning through adversarial optimization [5,6]. Finally, we compare the proposed lightweight approach with a large pre-trained wav2vec 2.0 model [9], providing a comprehensive analysis of performance and computational trade-offs.

II. Related Works

Early approaches to speech emotion recognition relied heavily on handcrafted acoustic features, such as pitch, energy, formants, and spectral descriptors, combined with classical machine learning classifiers. While these methods provided initial insights into emotional speech patterns, their reliance on manual feature engineering limited scalability and generalization.

With the rise of deep learning, end-to-end and feature-learning approaches have largely replaced traditional pipelines. CNN-based models operating on log-Mel spectrograms have demonstrated strong performance by learning local time–frequency patterns directly from data [3]. Recurrent architectures, including LSTMs and GRUs, have been widely adopted to capture temporal dynamics across speech segments [12]. Hybrid CNN–RNN architectures, commonly referred to

as CRNNs, combine the strengths of both paradigms and have become a standard baseline for SER tasks [4].

Domain adaptation techniques aim to reduce the discrepancy between training and testing data distributions. Among these, Domain-Adversarial Neural Networks (DANN) introduce an auxiliary domain classifier and a Gradient Reversal Layer (GRL) to encourage the learning of domain-invariant features [5,6]. DANN has been successfully applied in computer vision and speech-related tasks, but its systematic evaluation in cross-dataset SER scenarios remains relatively limited.

More recently, self-supervised pre-trained speech models such as wav2vec 2.0 have achieved state-of-the-art results across various speech processing tasks [9]. These models learn rich representations from large amounts of unlabeled audio and can be fine-tuned for downstream tasks. However, the extent to which pre-training alone addresses domain shift in SER remains an open research question.

III. Proposed Method

The proposed approach is based on a lightweight CRNN architecture extended with adversarial domain adaptation. The goal is to learn feature representations that are both discriminative for emotion classification and invariant across domains.

A. Datasets

Two widely used emotional speech datasets are employed in this study: IEMOCAP and RAVDESS. IEMOCAP serves as the source domain for supervised training, while RAVDESS is used as a target domain for cross-domain evaluation.

IEMOCAP consists of approximately 12 hours of dyadic interactions between pairs of speakers. The dataset is annotated with multiple emotional categories based on human judgments. In this work, five emotion classes are considered: angry, happy, sad, neutral, and frustrated. To ensure speaker-independent evaluation, the dataset is split by sessions, with Sessions 1–3 used for training, Session 4 for validation, and Session 5 for testing [1].

RAVDESS is a standardized acted emotional speech dataset containing recordings from multiple speakers expressing predefined emotional states [2]. Although RAVDESS shares nominal emotion labels with IEMOCAP, it differs substantially in recording conditions, speaking style, and emotional expression, making it suitable for evaluating cross-domain generalization.

All audio signals are resampled to a fixed sampling rate and converted to log-Mel spectrograms [3]. Fixed-length segments are obtained through padding or truncation to ensure consistent input dimensions across samples.

B. Architectures

- CRNN Feature Extractor
  The baseline model employs a Convolutional-Recurrent Neural Network. The input to the network is a log-Mel spectrogram represented as a two-dimensional time–frequency matrix. A stack of convolutional layers extracts local spectral features, followed by max-pooling operations that progressively reduce the temporal and frequency resolution [4].
  The output of the convolutional block is reshaped and passed to a bidirectional LSTM, which captures temporal dependencies in both forward and backward directions [12]. The recurrent output is aggregated across time using temporal pooling, producing a fixed-length representation for each speech segment.
  Dropout regularization was applied to the classifier and recurrent layers in order to mitigate overfitting during training [11].
- Domain-Adversarial Neural Network (DANN)
  To address domain shift, the CRNN is extended using a Domain-Adversarial Neural Network. In addition to the emotion classifier, a domain classifier is introduced to predict whether a sample originates from the source or target domain. A Gradient Reversal Layer (GRL) is placed between the feature extractor and the domain classifier [5,6].
  During backpropagation, the GRL multiplies the gradient by a negative scalar α, effectively reversing the optimization objective for the feature extractor [6]. This adversarial process encourages the feature extractor to learn representations that are informative for emotion classification while being uninformative for domain discrimination.
  The emotion classification loss is defined as the categorical cross-entropy:

$$L_{\text{cls}} = -\sum_{c} y_c \log(p_c) \tag{1}$$

  where y_c denotes the ground truth label and p_c the predicted probability for class c.
  The domain classification loss is defined similarly:

$$L_{\text{domain}} = -\sum_{d} y_d \log(q_d) \tag{2}$$

  where q_d denotes the predicted probability for domain d.
  The overall training objective is given by:

$$L_{\text{total}} = L_{\text{cls}} + \lambda\, L_{\text{domain}} \tag{3}$$

  where λ controls the trade-off between emotion discrimination and domain invariance [5].
- Evaluation Metrics
  Model performance is evaluated using accuracy, precision, recall, and F1-score [7]. Given the presence of class imbalance, F1-score is particularly important. Confusion matrices are also employed to provide detailed insight into class-wise behavior and error patterns beyond overall accuracy [8].
  The architectural design choices, training hyperparameters, and evaluation settings for all examined models are summarized in Table I.

TABLE I
MODEL ARCHITECTURES, HYPERPARAMETERS, AND PERFORMANCE

Configuration	Baseline CRNN	DANN (CRNN)	Wav2Vec2 + DANN
Parameters	∼2.2M	∼2.2M	∼95.2M (encoder ∼95M, heads ∼247K)
Epochs	20	100	10
Batch Size	16	16	4
Learning Rate	1e−3	5e−4	2e−5 enc., 3e−4 cls.
Optimizer	Adam	Adam	AdamW
Weight Decay	None	None	0.01
LR Scheduler	None	ReduceLROnPlateau	ReduceLROnPlateau
Gradient Clipping	None	None	max norm = 1.0
λ_{domain}	–	0.7	0.5
$\alpha_{\max}$	–	0.7	0.7
Classifier Hidden	256 → 128	256 → 128 → 64	256
Domain Head Hidden	–	64	64
IEMOCAP Accuracy	44%	45%	42%
RAVDESS Accuracy	14%	31%	27%

IV. EXPERIMENTS AND RESULTS

During training, data augmentation techniques such as time stretching, pitch shifting, and noise injection were applied exclusively to the source domain in order to improve generalization and robustness [10].

Experiments are conducted to evaluate both in-domain and cross-domain performance. Models are trained on the IEMOCAP dataset and evaluated on the IEMOCAP test split as well as the RAVDESS dataset.

Class-wise prediction behavior and cross-domain generalization effects are analyzed using row-normalized confusion matrices for all examined models and datasets (Figs. 1–6).

The confusion matrices of the baseline CRNN reveal a clear discrepancy between in-domain and cross-domain performance. While the model demonstrates balanced class predictions on IEMOCAP (Fig. 1), its performance collapses when evaluated on RAVDESS, with predictions heavily biased toward a dominant class (Fig. 2).

In contrast, the DANN-based model exhibits improved class balance and reduced prediction bias under cross-domain evaluation. As shown in Fig. 3 and Fig. 4, adversarial domain adaptation preserves source-domain performance while substantially improving generalization to the target domain.

The wav2vec 2.0-based model combined with DANN further improves overall recognition performance, particularly on the source domain, as illustrated in Fig. 5. However, although cross-domain predictions on RAVDESS are more balanced than the baseline, residual confusion between emotionally similar classes remains evident (Fig. 6).

A. Compute Environment

All experiments are conducted using a local workstation equipped with a CUDA-capable GPU with limited memory capacity. Lightweight CRNN-based models are trained efficiently within minutes, whereas fine-tuning the wav2vec 2.0 model requires significantly more computational resources and longer training times due to its large parameter count [9]. These constraints influence batch size selection and the number of training epochs.

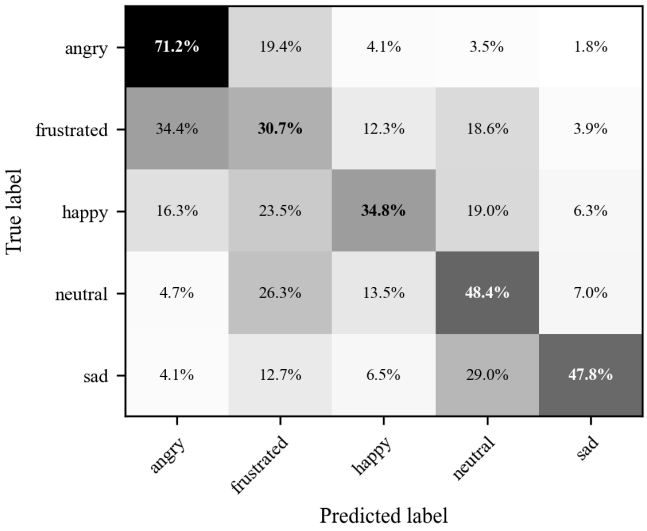

Fig. 1. Row-normalized confusion matrix of the Baseline CRNN on IEMOCAP.

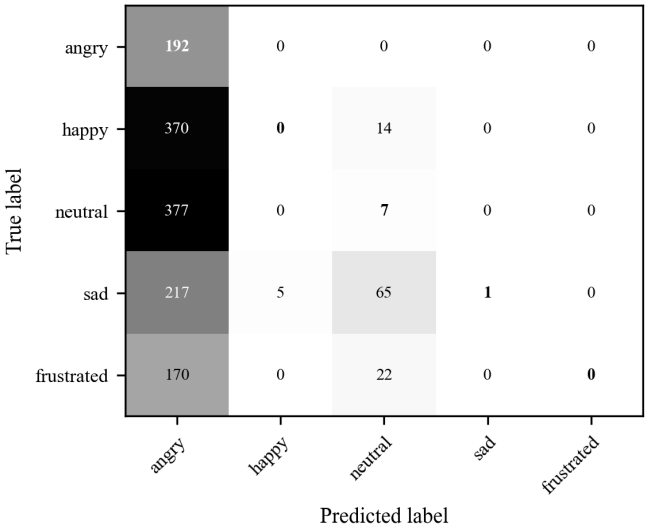

Fig. 2. Row-normalized confusion matrix of the Baseline CRNN on RAVDESS.

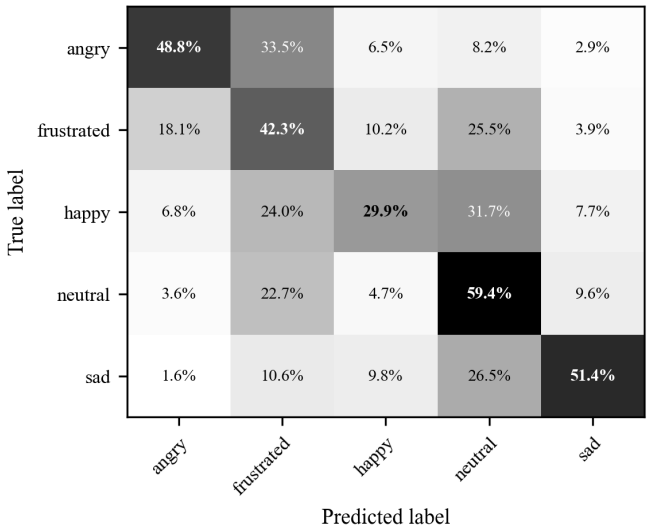

Fig. 3. Row-normalized confusion matrix of DANN (CRNN) on IEMOCAP.

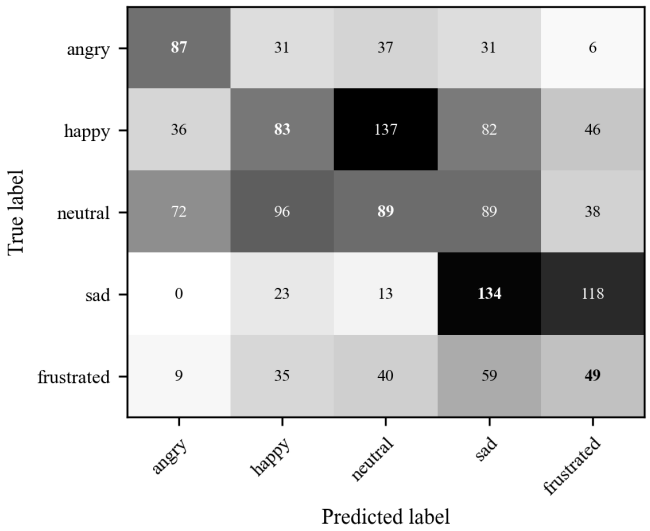

Fig. 4. Row-normalized confusion matrix of DANN (CRNN) on RAVDESS.

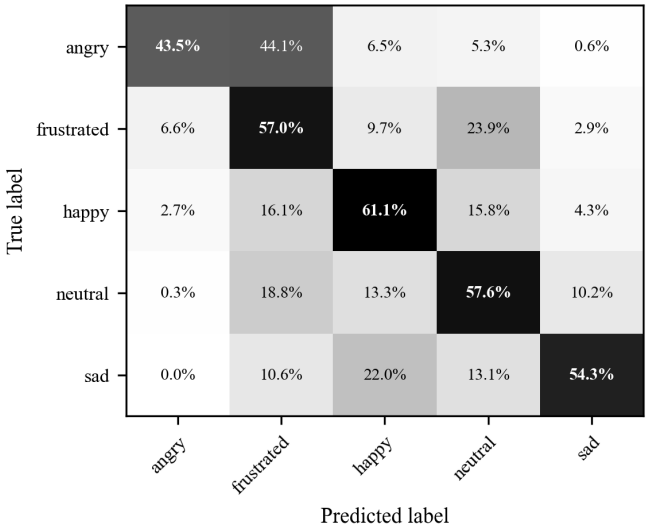

Fig. 5. Row-normalized confusion matrix of Wav2Vec2 + DANN on IEMOCAP.

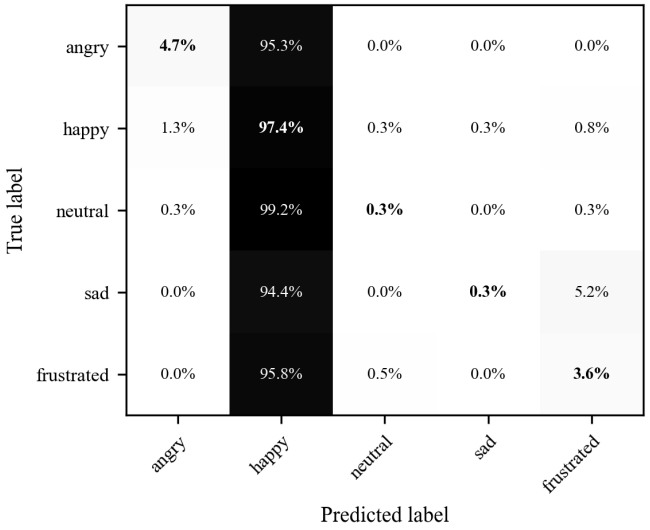

Fig. 6. Row-normalized confusion matrix of Wav2Vec2 + DANN on RAVDESS.

V. Discussion

The experimental results clearly demonstrate that domain shift constitutes a critical challenge for speech emotion recognition systems. Although the baseline CRNN achieves reasonable performance on the IEMOCAP test set, its severe performance degradation on RAVDESS confirms that high in-domain accuracy does not guarantee robust generalization across datasets with differing characteristics. This observation highlights the limitations of conventional supervised training when applied to emotionally annotated speech originating from heterogeneous domains.

The introduction of Domain-Adversarial Neural Networks substantially mitigates this issue by encouraging the learning of domain-invariant feature representations. The DANN-based model consistently improves cross-domain performance while preserving source-domain accuracy, indicating a favorable trade-off between emotion discriminability and domain invariance [5,6]. Analysis of the confusion matrices further reveals improved class balance on the target domain, suggesting that adversarial training reduces bias toward dominant emotion classes commonly observed under domain shift.

Comparisons with wav2vec 2.0 indicate that large self-supervised pre-trained models provide powerful speech representations and achieve competitive performance in both source and target domains [9]. However, the results also show that pre-training alone does not fully resolve domain shift, as explicit domain adaptation remains necessary to ensure robust cross-dataset generalization. Furthermore, the high computational cost and memory requirements associated with wav2vec-based models limit their practicality in real-time or resource-constrained settings.

In contrast, the lightweight CRNN-based DANN model offers a more efficient alternative, achieving comparable cross-domain improvements with significantly fewer parameters and reduced training time. This efficiency makes the proposed approach particularly attractive for deployment in applied settings, such as psychotherapy-oriented speech analysis, where scalability, interpretability, and computational feasibility are important considerations.

VI. Conclusion and Future Work

This paper presented a systematic investigation of domain shift and domain adaptation in speech emotion recognition. By extending a lightweight CRNN baseline with adversarial domain adaptation, we demonstrated that Domain-Adversarial Neural Networks significantly enhance cross-domain generalization while maintaining stable performance on the source domain. Experimental results across IEMOCAP and RAVDESS confirm that explicit domain adaptation mechanisms are essential for robust emotion recognition in heterogeneous speech data.

A comparative analysis with a pre-trained wav2vec 2.0 model further revealed important trade-offs between model capacity, computational cost, and generalization. While wav2vec-based approaches benefit from large-scale self-supervised pre-training, the proposed CRNN-based DANN model achieves competitive cross-domain performance with substantially lower computational requirements. This finding underscores the practical advantages of lightweight architectures combined with domain adaptation, particularly for applications requiring efficient training and deployment.

Future work will primarily focus on refining domain adaptation strategies for pre-trained speech models, including tighter integration of DANN mechanisms within wav2vec-based architectures [9]. Additional directions include multi-source domain adaptation, attention-based temporal aggregation to better capture emotionally salient speech segments, and the evaluation of the proposed framework on more naturalistic psychotherapy and clinical speech datasets. These extensions aim to further improve robustness and applicability in real-world affective computing scenarios.

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